Digital filters can be implemented in two ways, by *convolution* (also called *finite impulse response* or *FIR*) and by *recursion* (also called *infinite impulse response* or *IIR*). Filters carried out by convolution can have far better performance than filters using recursion, but execute much more slowly.

The next six chapters describe digital filters according to the classifications in Table 14-1. First, we will look at filters carried out by convolution. The *moving average* (Chapter 15) is used in the time domain, the *windowed-sinc* (Chapter 16) is used in the frequency domain, and *FIR custom* (Chapter 17) is used when something special is needed. To finish the discussion of FIR filters, Chapter 18 presents a technique called FFT convolution. This is an algorithm for increasing the speed of convolution, allowing FIR filters to execute faster.

Next, we look at recursive filters. The *single pole* recursive filter (Chapter 19) is used in the time domain, while the *Chebyshev* (Chapter 20) is used in the frequency domain. Recursive filters having a custom response are designed by *iterative techniques*. For this reason, we will delay their discussion until Chapter 26, where they will be presented with another type of iterative procedure: the neural network.

As shown in Table 14-1, *convolution* and *recursion* are rival techniques; you must use one or the other for a particular application. How do you choose? Chapter 21 presents a head-to-head comparison of the two, in both the time and frequency domains.